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Gaussian process-based Bayesian optimization for data-driven unit commitment

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ABSTRACT

Global efforts aiming to shift towards de-carbonization give rise to remarkable challenges for power systems and their operators. Modern power systems need to deal with the uncertain and volatile behavior of their components (especially, renewable energy generation); this necessitates the use of increased operating reserves. To ameliorate this expensive requirement, intelligent systems for determining appropriate unit commitment schedules have risen as a promising solution. This is especially the case for weak power systems with low dispatching flexibility and high dependency on imported fossil fuels. In this work, we introduce a radically new paradigm for addressing the optimal unit commitment problem, that is capable of accounting for the largely unaddressed challenge of the uncertain and volatile behavior of modern power systems. Our solution leverages widely adopted developments in the field of uncertainty-aware machine learning models, namely Bayesian optimization. This framework enables the effective discovery of the best possible configuration of a volatile system with uncertain and unknown dynamics, without the need of introducing restrictive prior assumptions. Based on appropriately selected acquisition function and Gaussian process regression, it constitutes a radically different from existing approaches, which heavily rely on heuristic approximations and do not allow to account for volatile behavioral patterns. On the contrary, it guarantees global optimum solutions in non-convex optimization tasks in the least possible number of trials. The demonstrated results show better performance in terms of total production cost and number of function evaluations, inspiring system operators to better schedule their power networks in the forthcoming, de-carbonized grids.

1. Introduction

Intelligent scheduling of power systems for seamless integration of intermittent renewable generation (RG) and plug-in electric vehicles (PHVs) constitutes a crucial solution in delivering future low carbon energy [1]. The variability and uncertainty caused by increased penetrations of RG must be properly considered in day-ahead unit commitment (UC), optimal power flow and even real-time economic dispatch (ED) problems. This is necessary in order to cope with operational issues such as insufficient ramping capabilities of thermal generating units, inadequate spinning reserve, transmission congestion and involuntarily customer demand interruptions [2]. System operators need to devise effective flexibility adequacy plans for their power systems so as to guarantee power balance and ensure feasible and economical operation over different time horizons and under different generational, environmental and technical constraints [3]. Typically, these constraints are captured by means of UC models developed to determine the commitment status and generation dispatch of various generating units; this is

Abbreviations: a_i, b_i, c_i, cost coefficients of unit i (\$/h, \$/MWh, \$/MW^{2}h); CSU_i, cold start-up cost of unit i (\$/h); $C_{SU,\,i}$, start-up cost of unit i (\$/h); D_t, data set; f, objective cost (\$/h); f^*, maximum performance rating; HSU_i, hot start-up cost of unit i (\$/h); J^*, relaxed total cost (\$); k, kernel function; MU_i, minimum up time of unit i (h); MD_i, minimum down time of unit i (h); N, number of participating generating units; P^t_{VRES}, variable output from RES during interval t (MW); $p(D|f)$, likelihood; $p(f)$, prior; $p(D)$, marginal likelihood; $p(f|D)$, posterior; P^t_i, power output of unit i during t (MW); $P^t_{max\text{-}VRES}$, maximum expected output from RES (MW); P^t_{netD}, net load demand during t (MW); $P_{i,}\,{}^{t}_{max}$ maximum capacity of unit i (MW); $P_{i,}\,{}^{t}_{min}$ minimum capacity of unit i (MW); q^*, corrected total cost (\$); RU^t_i, ramp-up capability of i (MW/h); RD^t_i, ramp-down capability of i (MW/h); RDG, relative duality gap (%); SR^t_{up}, upward spinning reserve (MW); SR^t_{down}, downward spinning reserve (MW); t, time interval; T, time horizon (intervals); TPC, total production cost (\$); U^t_i, on-off status of unit i during t (1-0); u_c, acquisition function; x_*, optimised variable vector; λ, Lagrange multiplier; σ, covariance.

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repeated for each interval of a short-term scheduling period (hours, days or a week).

The UC problem has commonly been interpreted as a large-scale, non-convex, mixed-integer and non-linear optimization problem, with a large number of complex constraints. The various approaches broadly used can be divided into mathematical, heuristic, *meta*-heuristic and hybrid. Ranging from classical mathematical to hybrid *meta*-heuristic, the most important methods are categorized as: Dynamic Programming, Priority List, Lagrangian Relaxation, Mixed-Integer Linear Programming, Quadratic Programming, Stochastic Programming, Branch-and-Bound, Simulated Annealing, Genetic Algorithm, Tabu Search, Ant Colony, Particle Swarm Optimization, Fuzzy Logic, Artificial Neural Network, Decision Tree and other hybrid techniques [4,5]. Mathematical optimization constitutes a more flexible approach allowing the constraints to easily be added. Priority-list schemes are fast and rely on the order of increasing operating cost to commit generation-units [6]. Dynamic programming suffers from the curse of dimensionality calling for some simplification techniques and opening a special field of hybridization. Due to its ability to overcome the difficulty of non-convexity and non-linearity of large-scale systems, a global optimum (if any exists) can be achieved by complete enumeration. Lagrangian relaxation is easily modified to model characteristics of specific utilities; it is capable of incorporating unit constraints separately and coupling constraints by just defining a respective Lagrange multiplier which is adjoined into the objective function of the relaxed problem [7]. Although it may be the most attractive for large systems, only a near optimal feasible solution can be expected. Another weakness regards the sensitivity problem which may cause unnecessary commitments of some units (e.g. identical generating units) [8].

On the other hand, heuristic methods are non-rigorous, empirical methods that make UC decisions based on pre-calculated priority list and incorporate the operating constraints heuristically. Although they are flexible (allowing practical operating constraints to be integrated) and require moderate computational memory and running time, the optimal solution cannot be guaranteed especially in large-scale power systems where even the magnitude of their sub-optimality cannot be estimated. Aiming to employ more rigorous methods generating more economical solutions, *meta*-heuristic approaches have been proposed in the expense of large computational efforts requirement. Both categories require great experience relating to the whole power systeḿs behaviour for their confident design and implementation. However, they include randomness and use stochastic approach in moving from one solution to another, compared to mathematical techniques which follow deterministic transition rules. In contrast to heuristic which are very specific and problem-dependent, *meta*-heuristics apply higher-level techniques aimed at problem-independent solutions. Consequently, they increase exponentially the problem dimensionality which in turns leads into unacceptable, in terms of convergence time, solutions.

Despite these advances, the recent expansion of intermittent RG and price-responsive demand participation has posed new challenges to the UC formulation and solution. Specifically, their uncertain and volatile behavior defies the deterministic nature of the conventional formulation of the UC problem as a non-linear optimization problem. Current attempts to attenuate the immensity of this problem have relied on a variety of solutions. Aiming to offer the flexibility needed in the presence of variable energy sources, the authors in [9] proposed the controlling of demand by electric water heaters incorporating a physically based aggregate model of the population in the day-ahead UC. Peak load shifting in municipal utilities has been achieved through a large controllable temperature range and the related work may be extended to consider other electric loads with energy storage such as electric vehicles, space heating and cooling loads. A similar work found in [10] proposes a daily UC that utilizes interruptible load to deal with the increased uncertainty due to wind power. Both load and wind are treated as uncertainty sources and are assumed to follow normal distribution and the UC problem is effectively solved by the adoption of a hybrid Lagrange Relaxation-Dynamic Programming algorithm. More recently, [11] introduced the terms of deterministic and stochastic unit commitment, with focus on the determination, allocation and deployment of reserves. The UC model formulated constitutes a Mixed Integer Linear Programming approach and is divided into two stages. The first stage involves the day-ahead scheduling (hourly deterministic unit commitment) and the second is based on real-time dispatch and UC which addresses the intra-hourly forecast errors (uncertainty) and system ramping requirements (variability).

This work constitutes an attempt to effectively address the inherent uncertainty that plagues the UC problem for RG systems, by leveraging the latest advances in machine learning. Our solution relies on the framework of Bayesian optimization (BO) [12]: This is a machine learning-driven optimization paradigm that uniquely allows to take into account the stochastic behavior of UC components, especially RG systems. This way, it is capable of generating reliable estimates of the magnitude of their expected variation; this is achieved by using a Bayesian regression technique that allows for uncertainty-aware prediction, namely Gaussian processes (GPs) [13]. Crucially, BO uses the GP-derived uncertainty estimates to guide an indicate iterative optimization process, which aims at obtaining the globally optimal UC setup in the least possible number of iterations (trials). In addition, another crucial advantage of BO consists in completely obviating the need for specifying an explicit functional form for the dependence between the optimized stochastic variable, y, and the observed independent variables, $\mathbf{x}$, $y = f(x)$. On the contrary, the functional form of this dependence is inferred from observed data, both historical and actively collected during the optimization process, in a non-parametric Bayesian fashion. Apparently, this paradigm offers the opportunity of discovering optima of unprecedented quality, as it ameliorates the reliance on assumptions on the functional form of $f(x)$ which might be far from the reality.

Note that, although there is a large literature about BO, most research works concentrate in optimizing either continuous or integer variables. Since the most complex and nonlinear computational models in fields such as behavioural, cognitive and computational neuroscience cannot be evaluated analytically, but require moderately expensive numerical approximations or simulations, Luigi Acerbi and Wei Ji Ma proposed BO to optimally fit these models [14]. Using GP as the posterior estimator, Javad Azimi, Ali Jalali and Xiaoli Z. Fern analysed BO in sequential and batch mode of evaluation to determine an effective experimental design according to the batch size [15]. A demonstration of the challenge of hyper-parameter optimization in large and multilayer models including stacked denoising autoencoders, deep belief networks (DBNs), convolutional networks and classifiers based on sophisticated feature extraction techniques is provided in [16]. Given an underlying metric between pairs of set elements, Garnett, Roman, Michael A. Osborne, and Stephen J. Roberts [17] introduced a natural metric for the task of selecting an optimal set of sensors by the predictive accuracy of the resulting sensor network, while Brochu, Eric, Tyson Brochu, and Nando de Freitas [18] employed the Bayesian technique to assist users in finding good parameter settings of a procedural fluid animation system in as few steps as possible.

BO has been also applied in combination with adaptive Monte Carlo for solving problems in physics, statistics and machine learning [19], as well as in benchmarks and novel applications to automatic information extraction [20]. The demonstrated results showed that BO forms a more efficient technique from which one can draw inspiration. Besides, there has not yet been a BO formulation that can address a combinatorial optimization problem such that of UC. In general, the task of combinatorial optimization has been treated as a sequential, model-based optimization for general algorithm configuration. Utilizing appropriate acquisition functions and GP regression, our solution exempts local minimum recommendations and allows for global exploration in the areas of minimum mean and/or higher uncertainty. This is a critical requirement for non-convex optimization tasks that are directly associated with recursive improvement until convergence. We provide strong experimental evidence regarding the efficacy and performance of our approach by considering a real-world scenario of a power system consisting of 10 thermal generating units and a 24-hourly residual power demand. In this context, we employed our proposed approach to achieve a system-wise UC schedule that ensures the minimum production cost, satisfying the underlying technical and operational constraints. Following, we assessed our solution to large-scale power systems consisting of up to 100 generating units in the presence of increased variable renewable energy sources.

The rest of the paper is organized as follows. In the following Section, we define the UC problem along with the main constraints that must be always satisfied. Section 3 describes the BO framework, as well as the GP model that it uses for inference purposes. In Section 4, we introduce the proposed novel methodology for uncertainty-aware, BO-based optimal UC. In Section 5, we perform an extensive experimental evaluation. We consider optimal UC in a real-world power network and we employ our approach to achieve a configuration that reliably minimizes the associated production cost. We saw that our approach completely outperforms the existing alternatives by a large margin, thus providing an extremely beneficial tool to system administrators. Finally, the conclusions are drawn in Section 6.

2. Unit commitment problem formulation

The generic decision process of UC problem can be formulated so as to minimize operational cost subject to both system-wide and unit-specific constraints, including system power balance (P^t_{netD}), spinning reserve (SR), generator capacity limits (P_{min} and P_{max}), minimum up-time (MU) and down-time (MD), ramp rate limits (RU and RD), generator status restrictions, plant crew constraints (C_r) and so on. Load curve serves as an input data to the problem while the output is the commitment status and generation dispatch of various generating units [21].

2.1. Objective function

In most approaches, the total production cost of a power system consisting of thermal units is mainly represented by the cost of fuel and start-up cost. Shut-down cost is a constant or at least time-independent value for each thermal unit, and thus it is usually considered 0 for all generating units. Mathematically, the objective is formulated as follows:

$$f = min\sum_{t=1}^{T}\sum_{i=1}^{N}\left[F_i(P^t_i) + \left(1 - U^{t-1}_i\right)C^t_{su,i}\right]U^t_i, \tag{1}$$

where U^t_i, determines the on–off states of generating units and P^t_i corresponds to the real power output of generator i during interval t. The thermal generation cost $F_i(P^t_i)$ (expressed by Equation (2)) depends on the production level of each unit and the fuel cost coefficients a_i, b_i and c_i derived from the convolution of the specific fuel cost and heat rate curve of each generating unit [22,23]. The start-up cost depends on the residual heat during the cooling hours of each offline generating unit and is approximated by Equation (3). It can vary from a small value (HSU – hot start-up) to a maximum cold-start value (CSU) according to the time the ith unit has been off-loaded (T_i). $t_{i,cold}$ and MD_i are the cold start time and minimum down time, respectively [24].

$$F_i\left(P^t_i\right) = a_i + b_i.P^t_i + c_i.\left(P^t_i\right)^2 \tag{2}$$

$$C^t_{su,i} = \begin{cases} HSU_i, if MD_i \le T_i \le MD_i + t_{i,cold} \\ CSU_i, if T_i + t_{i,cold} < MD_i \end{cases} \tag{3}$$

2.2. Constraints

The selected constraints that must be satisfied throughout the optimization process include:

1) System power balance. Considering the contribution of RG (P^t_{RG}), the total committed production must meet the residual, net load demand (P^t_{netD}) at each time-interval.

$$\sum_{i=1}^{N} U^t_i.P^t_i = P^t_{netD} \tag{4}$$

$$P^t_{netD} = P^t_D - P^t_{RG} \tag{5}$$

2) Spinning reserve. The power margins must be satisfied based on the maximum ($P^t_{i,\text{max-}cap}$) and minimum ($P^t_{i,\text{min-}cap}$) ramping capacity of each unit, taking into account the ramp rate limits. This way, SR^t_{up} is needed in order to recover a probable generation deficit, whereas SR^t_{down} must be guaranteed in the case of excessive renewable power output.

$$\sum_{i=1}^{N} U^t_i.P^t_{i,\max\_cap} \ge P^t_{netD} + SR^t_{up} \tag{6}$$

$$\sum_{i=1}^{N} U^t_i.P^t_{i,\min\_cap} \le P^t_{netD} - SR^t_{down} \tag{7}$$

3) Generator capacity limits. The maximum and minimum rated power that force the generating units to operate within their boundaries.

$$U^t_i.P^t_{i,\min} \le P^t_i \le U^t_i.P^t_{i,\max} \tag{8}$$

4) Minimum up (MU_i) and down times (MD_i). The minimum duration period needed before a generator can change its status. t_u and t_d represent the time a unit has started-up or shut-down, respectively.

$$U^t_i = 0 \rightarrow 1 if \sum_{t=td}^{t-1} U^t_i \ge MD_i \tag{9}$$

$$U^t_i = 1 \rightarrow 0 if \sum_{t=tu}^{t-1} U^t_i \ge MU_i \tag{10}$$

5) Ramp up/down rates. The rate of change of power output between adjacent hours is restricted by ramp-up (RU_i) and ramp-down (RD_i) rates regarding each generation unit [3].

$$P^t_i - P^{t-1}_i \le RU_i, \quad if \;\; generation \;\; increases \tag{11}$$

$$P^{t-1}_i - P^t_i \le RD_i, \quad if \;\; generation \;\; decreases \tag{12}$$

6) Unit status restrictions. Must run, must out and run at fixed-MW output constitute the three possible states of generating units.

$$\forall t \in \mathscr{T}\begin{cases} U^t_j = 1, \;\; if \;\; j^{th} \;\; unit \;\; must \;\; run \\ U^t_j = 0, \;\; if \;\; j^{th} \;\; unit \;\; must \;\; out \\ P^t_j = \begin{cases} 0 if \;\; U^t_j = 0 \\ P_{j,const.} if \;\; U^t_j = 1 \end{cases}, \;\; if \;\; j^{th} \;\; at \;\; fixed \end{cases} \tag{13}$$

7) Plant crew constraints (Cr^t). The number of operators available pertains the number of units that can simultaneously start-up (or shut-down) [22]. The inequality constraint is formulated as follows:

$$\sum_{i=1}^{N} U^t_i\left(1 - U^{t-1}_i\right) \le Cr^t, \forall t \in T \tag{14}$$

3. Mathematical framework

We consider the problem of finding the global minimizer x_* of a scalar objective function $f(x)$ over some bounded domain, typically $\mu : \mathscr{X} \rightarrow \mathbb{R}$, subject to a set of K constraints $c_1, \cdots, c_K$ as defined previously.

$$x_* = \underset{x \in X}{\text{argmin}} f(x) s.t. c_1(x) \le 0, \cdots, c_K(x) \le 0 \tag{15}$$

BO introduces the fundamental assumption that the functional form of the scalar objective $f(x)$ is unknown and it cannot be determined with certainty. On the contrary, it constitutes a stochastic function of the input $\times$, which can only be modelled as a random process. On the basis of this assumption, BO infers a Bayesian predictive model for $f(x)$, by making use of previous function evaluations. The predictive model typically used is a GP, which operates as a relatively inexpensive surrogate to guide the optimization algorithm towards regions that are promising (low GP mean) and/or unknown (high GP uncertainty). This procedure is performed on the basis of a local optimization rule which relies on a properly defined acquisition function [14].

Let us consider an existing set of measurement pairs $\{x_i, y_i | i = 1, \cdots, n\}$; this constitutes both the observed data points and the corresponding measured values of the sought stochastic function f. BO uses this set as the training data D of an underlying GP model used for predictive inference purposes. GP inference consists in assigning a prior, $p(f)$, over the function $f(x)$ and, combining it with the evidence from the data D, formulated under a likelihood assumption, $p(D|f)$, to get a posterior distribution, $p(f|D)$, over $f(x)$ according to Bayes' theorem.

$$p(f|D) = \frac{p(D|f)p(f)}{p(D)} \quad or \quad posterior = \frac{likelihood*prior}{marginal \ \ likelihood} \tag{16}$$

In more detail, given any finite collection of n points $X = \{x_i \in \mathscr{X} | i = 1, \cdots, n\}$, the value of f at these points is assumed to be jointly Gaussian with mean m:X$\mapsto\mathbb{R}$, where $m = (\mu(x_1), \cdots, \mu(x_n))^T$ and covariance matrix K:X $\times$ X$\mapsto\mathbb{R}$, where $K_{ij} = k(\mathbf{x}_i, \mathbf{x}_j)$ for $1 \leq i,j \leq n$. This is expressed in the form of the postulated prior imposed over $f(x)$ which yields $f \sim \mathscr{N}(m, K)$. Typically, the mean is taken as zero, yielding:

$$f|X \ \mathscr{N}(0, K) \ \ or \ \ \log p(f|X) = -\frac{1}{2}f^T K^{-1} f - \frac{1}{2}\log|K| - \frac{n}{2}\log 2\pi \tag{17}$$

Turning to the selection of the covariance function, we typically utilize kernel functions that ensure its positive semi-definiteness. There is a large variety of alternatives that are typically used, including the RBF kernel, as well as Matern class kernels [25]. For instance, in this work we will make use of the Adaptive Relevance Determination (ARD) Matern 5/2 kernel, which reads:

$$k(x_i, x_j) = \sigma_f^2 \left(1 + \sqrt{5}\, r + \frac{5}{3} r^2\right) exp\left(-\sqrt{5}\, r\right) \tag{18}$$

$$r = \sqrt{\sum_{m=1}^{D} \frac{(x_{im} - x_{jm})^2}{\sigma_m^2}} \tag{19}$$

The benefit of this type of kernel lies on the fact that it puts different trainable weights, σ_m^2, on the components of the observed vectors x. This way, it learns which observed features are the most important for modelling purposes.

On the other hand, to allow for realistic modelling, we take into account that the GP model can only generate noisy predictions, $f(x)$, of the dependent variable y. To encapsulate this assumption, we postulate an additive independent and identically distributed Gaussian noise model, ε, with zero mean and variance σ_n^2. Under this formulation, we obtain a likelihood function for the model which reads $y|f \ \mathscr{N}(f, \sigma_n^2 I)$.

Based on the introduced prior and likelihood assumptions, and using Bayes' rule, it can be shown that the marginal likelihood of the model reads:

$$\log p(y|X) = -\frac{1}{2} y^T \left(K + \sigma_n^2 I\right)^{-1} y - \frac{1}{2}\log\left|K + \sigma_n^2 I\right| - \frac{n}{2}\log 2\pi \ \ or \ \ y|X \ \mathscr{N}\left(0, K + \sigma_n^2 I\right) \tag{20}$$

Optimization of this expression can be performed to estimate the appropriate values of the trainable hyperparameters of the employed kernel functions, for instance the set of σ_m^2 hyperparameters of the ARD Matern 5/2 kernel, as well as the noise variance hyperparameter $\sigma_n^2$of the likelihood of the model.

Once model training by optimizing the marginal likelihood (19) is performed, using a data set $D_t = \{x_{1:t}, y_{1:t}\}$, BO employs an acquisition function $u(x|D_t)$ to decide what point x_{t+1} to consider next, as it seeks to find the global optimum of the modelled stochastic function $f(x)$. Considering a minimization setup for $f(x)$, a point x_{t+1} yielding a high acquisition value should correspond to low posterior expected values of $f(x)$ inferred by the trained GP, or predictions with high uncertainty, as expressed in the form of the obtained predictive posterior variance.

The notion of expected improvement offers the most typically used formulation of the acquisition function. We consider [26]:

$$x_{t+1} = \underset{x \in X}{\operatorname{argmin}} u(x) \tag{21}$$

such that:

$$\begin{aligned} x_{t+1} &= \underset{x_t \in x_{1:t}}{\operatorname{argmin}} \mathbb{E}(\|f_{t+1}(x) - f(x_*)\| | D_t) \\ &= \underset{x_t \in x_{1:t}}{\operatorname{argmin}} \int \|f_{t+1}(x) - f(x_*)\| p(f_{t+1}|D_t) df_{t+1} \end{aligned} \tag{22}$$

In this expression, the predictive posterior $p(f_{t+1}|D_t)$ is obtained by making use of Bayes' rule and the likelihood and prior assumptions of the employed GP model. Specifically, the joint distribution of the observed target values and the function values $f_{t+1} = f(x_{t+1})$ at the test location x_{t+1} under the prior can be written as:

$$\begin{bmatrix} y_{1:t} \\ f_{t+1} \end{bmatrix} \mathscr{N}\left(0, \begin{bmatrix} \boldsymbol{K} + \sigma_n^2 \boldsymbol{I} & \boldsymbol{k} \\ \boldsymbol{k}^T & k(x_{t+1}, x_{t+1}) \end{bmatrix}\right) \tag{23}$$

where

$$\boldsymbol{K} = \begin{bmatrix} k(x_1, x_1) & \cdots & k(x_1, x_t) \\ \vdots & \ddots & \vdots \\ k(x_t, x_1) & \cdots & k(x_t, x_t) \end{bmatrix} \tag{24}$$

and

$$\boldsymbol{k} = \begin{bmatrix} k(x_{t+1}, x_1) & k(x_{t+1}, x_1) & \cdots & k(x_{t+1}, x_t) \end{bmatrix} \tag{25}$$

This yields the predictive distribution:

$$p(y_{t+1}|D_{1:t}, x_{t+1}) \ \mathscr{N}\left(m_t(x_{t+1}), \sigma_t^2(x_{t+1}) + \sigma_n^2\right) \tag{26}$$

and the sufficient statistics:

$$m_t(x_{t+1}) = \boldsymbol{k}^T \left[\boldsymbol{K} + \sigma_n^2 \boldsymbol{I}\right]^{-1} y_{1:t} \tag{27}$$

$$\sigma_t^2(x_{t+1}) = k(x_{t+1}, x_{t+1}) - \boldsymbol{k}^T \left[\boldsymbol{K} + \sigma_n^2 \boldsymbol{I}\right]^{-1} \boldsymbol{k} \tag{28}$$

Finally, in most practical optimization problems, the observation space does not cover the whole domain of real numbers; some parts of it are excluded due to imposed constraints. Therefore, the BO algorithm must be prohibited from picking such points when applying the expected improvement rule. To ensure this, we can modify the original expected improvement rule, weighting it with the probability of the constraints being satisfied [27,28].

$$u_C(x|D_t) = u(x|D_t) \prod_{k=1}^{K} p(c_k(x) \leq 0 | D_t) \tag{29}$$

Once observing the output of each query of the objective, the model is updated to produce a more informative posterior over the space of objective functions, the acquisition is recomputed, and a new input is chosen for evaluation. After n queries, the algorithm makes a final recommendation $x_{n\star}$, providing the algorithm's best estimate and completing the Bayesian optimization loop [29].

4. Methodology

Inspired from the previously described advances, in this paper we propose a radically novel paradigm for addressing the UC challenge. Our vision is: 1) to ameliorate the need of heuristically specifying a functional form for the fuel cost function, which constitutes a core part of the optimized objective function in conventional formulations of the UC problem, and may undermine the obtained outcome if it deviates from reality; and 2) allow to discover quality solutions to the UC problem after only a limited number of function evaluations, which is of utmost importance in real-world settings where scheduling decisions may need to be made within extremely limited time-windows.

To this end, our proposed approach utilizes the latest advances in Bayesian optimization under prespecified constraints, summarized in Section 2. Specifically, we formulate the UC problem as a BO task whereby the stochastic dependent variable, y, represents the total production cost of the system, while the independent variables, $\mathbf{x}$, comprise the optimizable variables that the algorithm needs to specify, namely the status of each unit (U_i^t) and the power output of each generating unit (P_i^t), in order to satisfy the exogenous variable, namely the net load demand (P_{netD}^t). Further, we impose a number of proper constraints to ensure the feasibility of the discovered solutions, both at the individual unit level and the system level. This way, we establish a novel inferential paradigm, which operates in a completely data-driven fashion to discover the system dynamics and utilizes this outcome to effect the optimization task.

In our setup, the power output at unit level is treated as a real number. On the other hand, the unit status constitutes an integer, binary variable. Under this scheme, it is easy to ensure satisfaction of the imposed status restrictions, for instance by excluding from the optimization the state variables of the units that *must run*, the power output variables of the units that run at a *fixed-MW output*, as well as both these two types of variables in cases of units that must be out.

Turning to the specification of the imposed constraints, we consider: 1) the predefined unit bounds; 2) a set of coupled constraints, which include the power balance constraint, as well as the spinning reserve requirements; 3) the plant crew constraint, defined in Equation (14); and 4) the MU/MD and RU/RD constraints, given by Equations 9–10 and 11–12, respectively.

5. Experimental evaluation

In this Section, we perform a thorough experimental assessment of our proposed solution, adopting a common experimental setup, described for instance in [30]. In the first part of our experimental study, we provide the validation of BO approach when applied in constrained UC tasks. We consider a power system comprised by 10 generators that are able to serve a total demand in the range of 660 MW and 2493 MW. Their properties are summarized in Table 1. For the purposes of this study, we treat the power demand as an input variable defined over daily horizons of 24-hourly intervals, as depicted in Table 2. The second part deals with an isolated power system consisting of 20 identical generating units in combination with intermittent renewable sources. Based on different real-world scenarios we assess the performance of the proposed solution compared with widely studied alternatives. All simulations were realized with MATLAB (R2018a) software running on 64bit Windows-10 and ASUS laptop machine with 2.4 GHz intel-core i7 processor (model 4700HQ) and 6 GB of RAM.

Table 2
Net load demand (MW) for 24 hourly intervals.

Hour	Load	Hour	Load	Hour	Load
1	700	9	1300	17	1000
2	750	10	1400	18	1100
3	800	11	1450	19	1200
4	850	12	1500	20	1400
5	950	13	1400	21	1300
6	1000	14	1300	22	1100
7	1150	15	1200	23	900
8	1200	16	1050	24	800

5.1. Method validation

We have implemented our BO paradigm in MATLAB 2018a [31]. To initialize the Bayesian optimization algorithm, we generate an initial guess of the independent variable values in x_0. These comprise an initialization of the UC, selected by considering that all generating units operate at their mean power output. Fig. 1 constitutes an example of the regression model used, along with its respective functions drawn at random from the GP prior and the posterior. This initialization strategy essentially reflects a random search UC policy, commonly used by transmission system operators.

To render our experimental setup more realistic, we introduce an additional constraint which is of major importance for system operators, but is omitted from the experimental setup of [30]: We consider a spinning reserve requirement, and set it at 50 MW, which is a standard practice in low-inertia and isolated power systems. Fig. 2 illustrates the outcome of the objective function evaluations at the determined candidate solutions over the performed algorithm iterations. For exposition purposes, we depict the solutions pertaining to the twenty-four hours. As we observe, for each hourly interval, the adopted BO process converges consistently. Characteristically, the first solutions, obtained through algorithm initialization, yield poor values; however, the performance improves fast as the BO algorithm proceeds.

In order to obtain some comparative results, we also evaluate some popular alternatives under the same experimental scenario. Specifically, we consider: 1) Dynamic Programming, which constitutes a standard solution in the related literature [5]; 2) the Lagrange Relaxation-based approach, presented in [7]; and 3) a standard Genetic Algorithm implementation, as provided in the GA toolbox of MATLAB and also considered in [32]. The algorithms relating to the Dynamic Programming approach and Lagrange relaxation method have been developed, following the conceptual formulation and logic consistency presented in the respective literature. The obtained results are provided in Table 3. Specifically, in this Table we provide the comparative results pertaining

Table 1
Characteristics of the thermal generating units.

Unit i	a ($/MW^{2}h)	b ($/MWh)	c ($/h)	SU ($)	Pmin (MW)	Pmax (MW)	RU & RD (MW/h)	MU & MD (h)
1[*]	0.00048	16.19	1000	4500	225	682.5	405	8
2	0.00031	17.26	970	5000	225	682.5	405	8
3	0.00200	16.60	700	550	30	195	54	5
4	0.00211	16.50	680	560	30	195	54	5
5	0.00398	19.70	450	900	37.5	243	67.5	6
6	0.00712	22.26	370	170	30	120	54	3
7	0.00079	27.74	480	260	37.5	127.5	67.5	3
8	0.00413	25.92	660	30	15	82.5	27	1
9	0.00222	27.27	665	30	15	82.5	27	1
10	0.00173	27.79	670	30	15	82.5	27	1

[*] Generating Unit *1* is constantly in must-run mode.

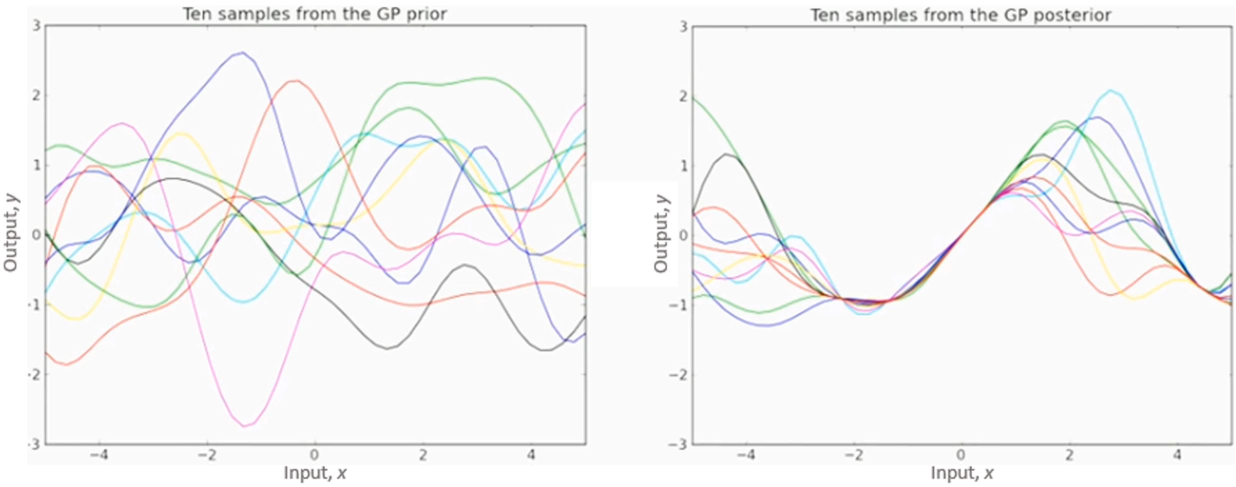


Fig. 1. Example of Gaussian process regression.

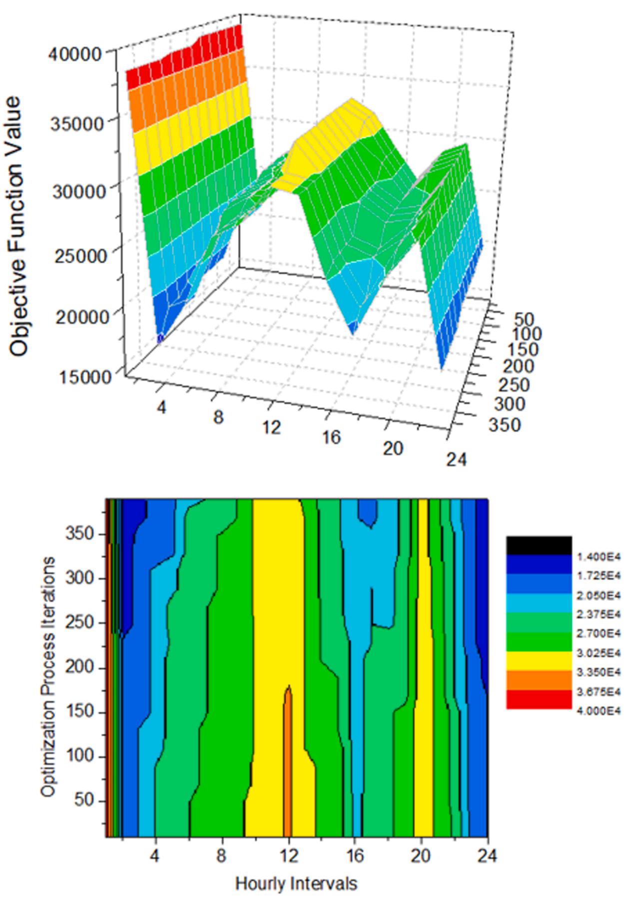


Fig. 2. Function evaluations of our proposed approach pertaining to the first twenty-four hourly intervals.

to the two key aspects that characterize an optimization process: 1) the number of attempts (function evaluations) required for the algorithm to converge (converge to a minimum); and 2) the eventually obtained optimal (cost) value.

According to the results, our proposed approach yields the best optimal solution, which decreases the minimum achieved TPC by $90 K. In contrast to Dynamic Programming which relies on complete enumeration of all the feasible states, Lagrange Relaxation offers the ability to decompose the problem into a master and a sub-problem,

Table 3
Comparative results.

Method	Number of Function Evaluations	Minimum TPC achieved
Proposed Approach	7840	504803.4
Lagrangian Relaxation	11,520	588217.2
Genetic Algorithm	24,000	596217.3
Dynamic Programming	$1.73\text{x}10^{72}$	504803.4

*TPC = Total Production Cost ($).

namely the UC and ED respectively. Based on dual optimization, the system-wide constraints linking the different sub-problems are embedded through a mathematical and implicit manner in order to solve the UC and whilst improving the lower bound at each iteration [8]. In this regard, the exploration space is limited due to the sensitivity of Lagrange multipliers, while the systematic adjustment relating to the update-rule optimization is inversely proportional to the execution time. This justifies the improvement in relation to BO, which refers to a greater than $83 K TPC. For completeness sake, in Fig. 3 we illustrate the optimal scheduling results of our approach over all the generating units, for the twenty-four hourly intervals.

It is worth noting that data-driven Bayesian optimization achieves global optimum solution in the least possible trials. When compared with Genetic algorithm, one can be seen is that the worst performance arises from the fact that heuristic approaches require consecutive tuning to find the optimum point within the produced population. In such a

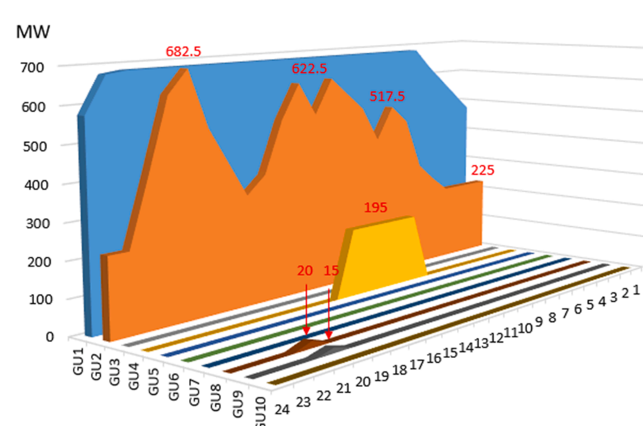


Fig. 3. Optimal UC schedule obtained by our approach.

non-convex optimization problem, the stochastic nature of selection processes may lead to sub-optimal recommendations. A further parameter that affects the optimization process is the convergence criterion used. Moving towards the regions with the minimum mean or maximum uncertainty, Gaussian process based BO utilizes an acquisition function which maximally improves the expected dependent-variable value. On the contrary, GA makes use of a no-change-in-the-solution rule for a pre-specified number of generations and thus, it is trapped into local optima concluding into poor, suboptimal recommendations.

The termination criteria relating to the rest of the examined methods vary according to the system requirements. Toward this goal, we set the execution time being lower than the time-horizon of each examined scenario for Dynamic Programming approach. As the fastest method, the trials for LR are limited by a minimum value of duality gap. In this regard, we define this value at the minimum 1.10^{-6}, which constitutes a quite low bound for the majority of dual-optimization applications. Utilizing the readily available function of genetic algorithm in MATLAB, we make use of the recommended parameter settings in terms of maximum number of generations ($\mathrm{maxG} = 100.\mathrm{nvars}$) and population size ($popS = 100$), where nvars is the number of integer and continuous variables. In our analysis, all constraint tolerances are set to 1.10^{-3}.

5.2. Real-world case-study

We extend our experiments by considering an isolated power system characterised by identical generating units and renewable energy sources, such as solar, wind and biomass. Biomass is considered as a firm import, whereas solar and wind are treated as variable sources requiring increased spinning reserves. This generates the need to define a radically different formulation for the spinning reserve, in order to account for the volatile behaviour of both variable sources and electricity demand. In addition, to maintain the *N*-1 criterion for isolated power systems, our formulation must also include the proportion of generation loss due to an unexpected unit failure. Consequently, the required spinning upward reserve for each time-interval can be calculated as:

$$SR_{up}^{t} = \max\{SR_1^t, SR_2^t\}, \tag{30}$$

where

$$SR_1^t = 6\% P_{netD}^t + 100\% P_{VRES}^t \tag{31}$$

and

$$P_{VRES}^t = P_{solar}^t + P_{wind}^t \tag{32}$$

SR_2^t represents the maximum capacity of the second biggest, in terms of power rating, generating unit being on-line. On the other hand, the downward portion of spinning reserve is restricted by the instantaneous maximum value $P_{max-RES}^t$ for each respective technology (e.g. for PV generators this value corresponds to the cloud-free production). Based on the projected values P_{VRES}^t, SR_{down}^t can be obtained as:

$$SR_{down}^t = P_{max-RES}^t - P_{VRES}^t \tag{33}$$

The necessary information regarding the identical generating units is tabulated in Table 4, while the corresponding weekly electricity demand is given in half-hour intervals in Fig. 4 along with the RG contribution.

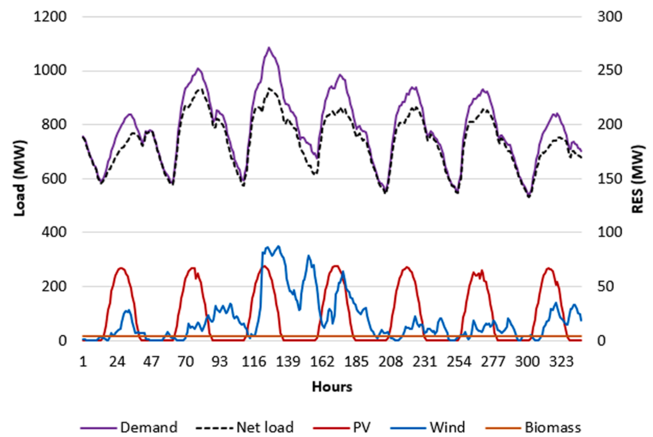


Fig. 4. Power demand versus RG contribution for a typical week in summer.

The necessity of robust formulations such that of (30) can be perceived with the aid of Fig. 5 and Fig. 6. The instantaneous maximum value is estimated by an average interpolation of the hourly summed power output from wind and PV generators illustrated in Fig. 5, whereas Fig. 6 shows their actual contribution per month.

Based on these characteristics and in order to assess our solution in large-scale power systems facing emerging electricity needs, we model three further generation profiles (concerning 40, 80 and 100 generators) to contribute in different scenarios. For simulation purposes, we consider 336 half-hour intervals to allow for the ramp-up and ramp-down constraints to take place. As expected, dynamic programming (DP) fails to deliver solutions at reasonable time-intervals. This is due to the increased combinations (reflected as objective function evaluations) that must be computed. For instance, in our first scenario this number reaches at least $(2^{20}-1)^{336}$ function evaluations. Since DP would offer a convergence in greater than one-week elapsed time, we exclude this method from the potential candidates.

The second case study accounts for a raise in power demand due to increased EVs. This necessitates a proportional increase in renewable energy contribution so that their share in final, useful energy to be maintained. In the third case, we strengthen the power requirements and RG penetration taking into account the upcoming trends of energy electrification, including heating and cooling sectors. Finally, we assume an interconnected, large-scale power system which allows for bulk integration of renewable energy in a fully liberalized electricity market environment. We provide the multiplication indices describing the mentioned scenarios in a separate table (Table 5).

Applying our approach, we observe during the first scenario that a total fuel cost of \$1.15 M can be achieved. It constitutes the least value following by Genetic Algorithm and Lagrange Relaxation with costs of 7.61 and 9.3 million dollars. This verify our beliefs that BO overrides in dealing with identical generating units. Fig. 7 demonstrates the optimal allocation across the 20 generating units during the first, middle and end part of the examined week.

The sensitivity of consecutive commitment and de-commitment of the identical generators during the Lagrange-Relaxation (LR) optimi-

Table 4
Characteristics of the identical generating units.

Unit i	a ($/MW^{2}h)	b ($/MWh)	c ($/h)	SU ($)	Pmin (MW)	Pmax (MW)	MU (h)	MD (h)	RU (MW/h)	RD (MW/h)
1–4	0.213034	67.84	949.05	104	4	37	0.5	3	150	150
5–10	0.171574	37.71	608.64	5786	30	58	2	8	60	30
11–13	0.013049	37.78	93.93	66	8.75	17	1	2	30	30
14–16	0.265645	31.36	113.93	66	14.5	17	1	2	30	30
17–18*	0.040193	34.41	750.27	9200	66	124	12	8	63	63
19–20	0.039537	43.19	2476.76	208	66	216	8	6	180	180

*Generating Unit *18* is constantly in must-run mode.

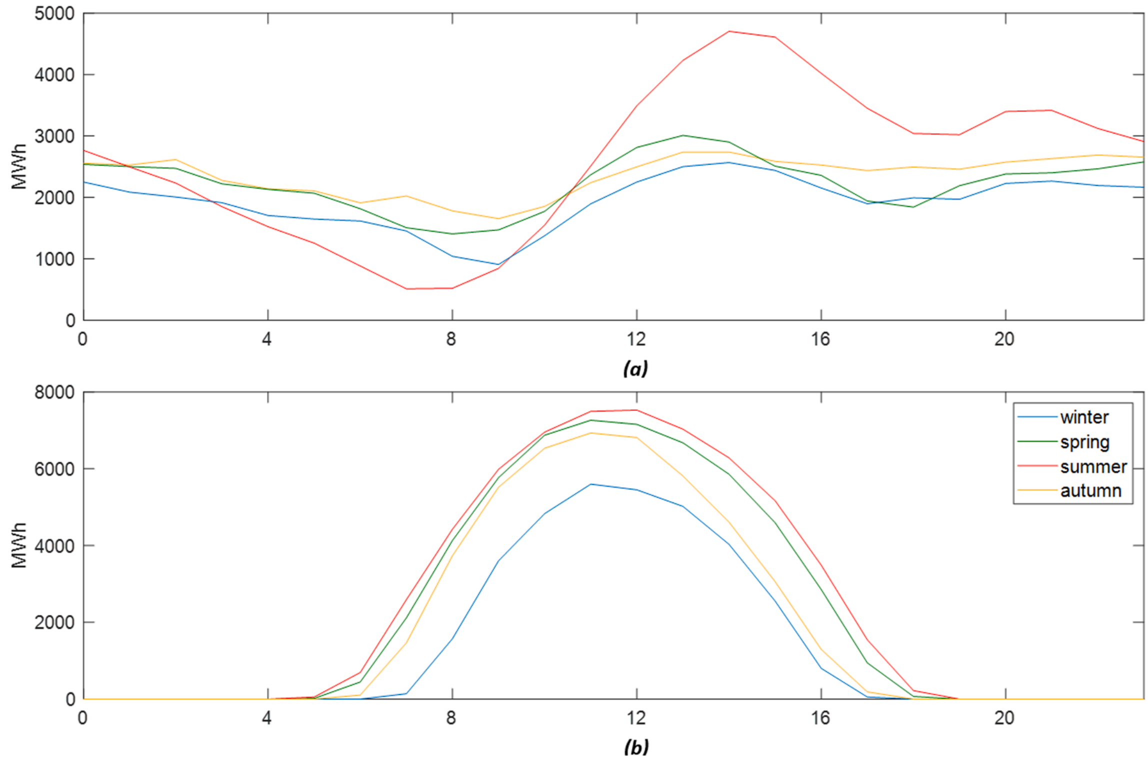


Fig. 5. Hourly summed contribution of *(a)* wind and *(b)* PV per season.

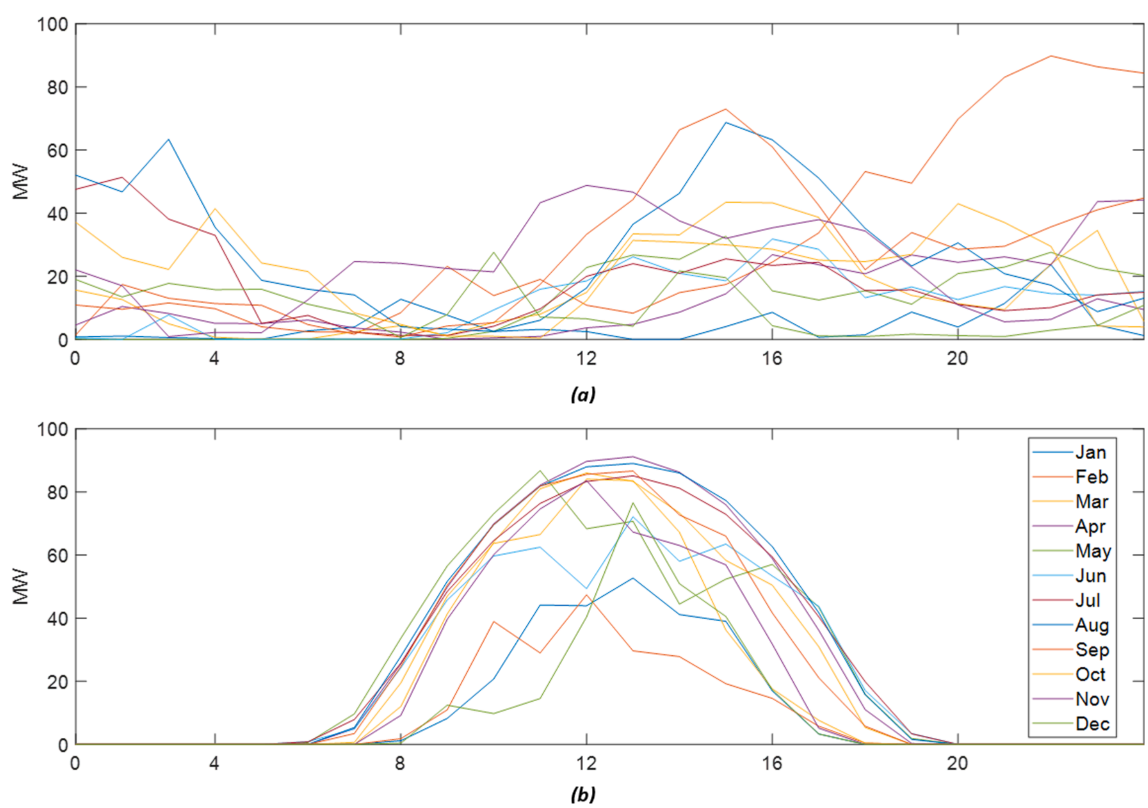


Fig. 6. Actual total power output of *(a)* wind farms and *(b)* PV parks per month.

zation process is translated as increasing start-up cost violating the convergence. Inherently, LR utilizes the value of the relaxed objective cost function to compare it with the dual, corrected cost and drive the optimization. As a result, the unnecessary commitments and decommitments of identical units lead to dual gap oscillations according to Equation (34) [33].

Table 5
Multiplication indices of the test case studies.

Scenario	Number of Generators	Load Demand	Biomass	PV	Wind
1	20	x1	x1	x1	x1
2	40	x2	x1	x2	x1.5
3	80	x4	x1	x4	x3
4	100	x5	x10	x5	x3.75

$$RDG = \frac{J^* - q^*}{q^*} \tag{34}$$

J^* represents the total cost (for the N generating units during the T time periods) estimated by the relaxed Lagrange multiplier (λ) while q^* is the total cost given by the corrected λ values used in economic dispatch. The over-scheduled generation and the associated increased start-up and production costs can be seen in Fig. 8. Based on their incremental generating cost, LR allows the participating units to consecutively be added until the power balance requirement is fulfilled. Although this mechanism constitutes a traceable pathway to convergence, it fails in concluding to an optimum solution when treating generators with identical heat-rate coefficients. As a result, the varying lambda value forces the identical units to start-up and contribute as a group rather than a single, separate unit which in turns increase the total production cost.

On the other hand, Genetic Algorithms completely avoid the undesirable oscillations by simply generating population-based solutions and keeping the fittest principle ones. However, the resulting recommendations may stand out from real optimum solutions; this can be seen approximately at the beginning of each day when the production from photovoltaics rises, forcing the residual load demand to fall evenly. During these intervals the spinning-reserve requirements become superior while the residual load leads the optimization to de-commit some generating units. The consequence of such a contradiction is a call for several smaller units to get on-line, in the expense of increased total start-up cost.

The performance of BO in combinatorial optimization tasks greatly improves by expanding the exploration space in terms of generating-units number and RG contribution. In addition, the half-hour intervals allow the constraints of ramp-up and ramp-down to take place restricting the feasible-solution range. Indeed, greater improvement in total production cost is observed in the following case studies. Fig. 9 depicts the rest scenarios of 40, 80 and 100 generating units respectively.

The total start-up cost expenses concerning GA still occur through Scenario 2, where the contribution of PV energy was doubled. Here the contradiction takes place both at the rising and falling of photovoltaic generation, since greater and more expensive to start-up units come to replace the production of smaller generators. LR still possesses the highest total production cost ($20.98 M), dominated by consecutive commitments and decommitments. Increasing the number of identical generating units of each group leads to involuntary start-ups and consequent uneconomic dispatch. In addition, the consecutive alternation of the status (on/off) of the groups until the demand satisfaction, deteriorates the overall optimization performance as it requires much more computational effort to converge to feasible recommendations.

The LR approach occupies the second place in the ranking when the contribution of wind farms starts growing. This is realized in the third and fourth scenarios and the phenomenon is repeated quite more often, according to Fig. 9 (b) and (c). Consequently, GA gives its position to LR which still oscillates around a relatively constant mean value, while our proposed approach remains smooth and flatter as the number of available generators, along with the renewable contribution, grow. For comparison purposes all simulation results relating to 10, 20, 40, 80 and 100 generating units case studies are included in Table 6.

A further critical characteristic affecting the appropriate selection of an optimization method is the number of function evaluations. It is proportional to the generating-unit number (N) and time intervals (T) as well as to the overall optimization procedure. For instance, DP is based on complete enumeration and thus all combinations need to be included according to Equation (35).

$$n = \left(2^N - 1\right)^T \tag{35}$$

This requires bulk capacity which it may not exist in cases where T is greater than 24. Furthermore, for $N \leq 10$, an alternative approximation can be used when RG does not violate the overall constraints. We utilize this advantage to compute the total production cost (*TPC*) in our first case study. Thus, we perform a process by taking into account $(2^{10} - 1)24 = 24552$ function evaluations rather than $\left(2^{10} - 1\right)^{24} = 1.73x10^{72}$ objective values.

Turning to the Lagrange Relaxation-based decomposition

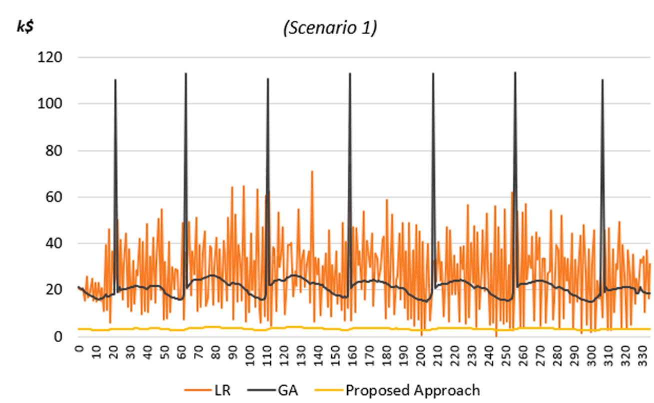


Fig. 8. Production cost pertaining the 336 half-hours of the examined week.

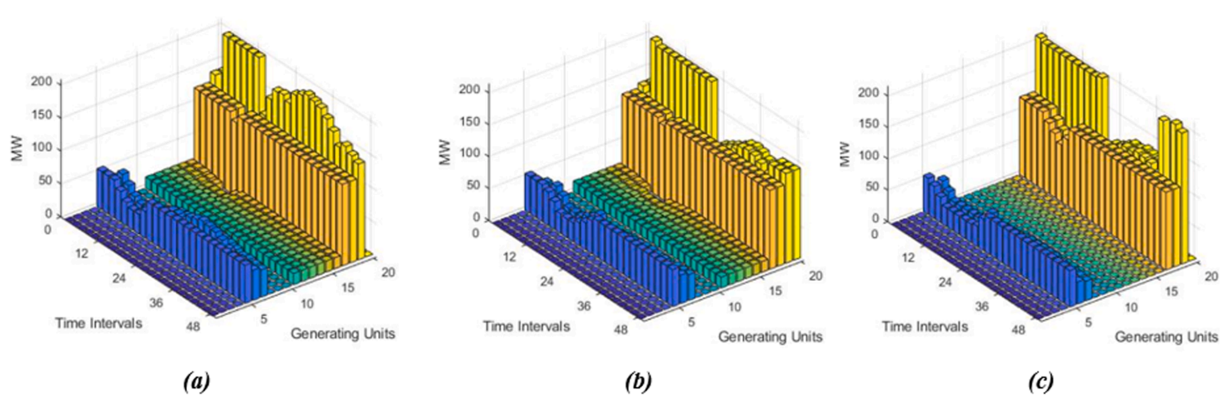
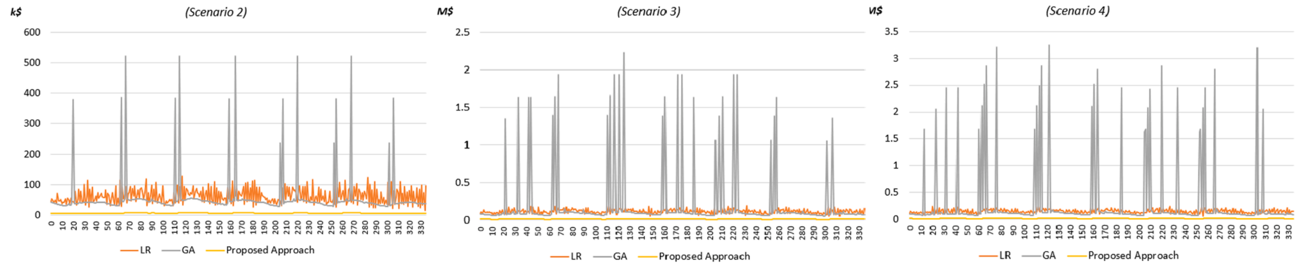


Fig. 7. Optimal economic dispatch of our proposed approach pertaining the 48 half-hours of (a) Monday, (b) Wednesday and (c) Sunday.



(a) (b) (c)

Fig. 9. Total production cost pertaining the 336 half-hours of the examined week, pertaining (a) 40, (b) 80 and (c) 100 generating units.

Table 6
Total number of function evaluations.

Scenario	Number of Generators	Dynamic Programming	Genetic Algorithm	Lagrange Relaxation	Proposed Approach
1	20	$(2^{20}-1)^{336} \to \infty$	336,000	7,200,000	7680
2	40	∞	672,000	201,600,000	100,800
3	80	∞	1,344,000	403,200,000	201,600
4	100	∞	1,680,000	806,400,000	403,200

approaches, these depend on the maximum number of iterations needed to get a minimum value, with no further significant improvement. The maximum number of function evaluations is calculated by multiplying the performed iterations by the number of available generators (must-out generating units are excluded) and time intervals (e.g. 24 or 336). In a similar manner, GAs impose a function evaluation sequence according to the population selected within the process. In this work, we chose a population of 300 optimal solutions to extract the best possible *TPC* value at a reasonable computational cost.

To give a comprehensive overview of the performance of our proposed solution based on GP inference and BO optimization, we normalized the results in the five distinguished categories. The normalization was based on Equation (36) as follows:

$$Rank(f) = 10\frac{f^{*} - f_i}{f^{*}} \tag{36}$$

f^{*} represents the maximum value between all candidates considered in each case study, whereas f_i is the value of each candidate. In this regard, the lowest values for total production cost, number of start-ups, total start-up cost, number of function evaluations and elapsed time correspond to higher rankings (e.g. 10). A less beneficial approach gives lower rank estimates while the worst would approximate zero.

The last feature which ultimately affects the selection of the best-fit optimization method refers to elapsed time. As can be observed, this characteristic sets DP out of choice while advantaging LR. Our solution stands in between and constitutes the best solution, offering a method able to drive optimization into global explorations of minimum objective function value. The superiority of our approach lies in the fact that it enables optimal scheduling of identical generating units in large-scale, interconnected power systems in the presence of volatile renewable sources with *TPC* in the order of \$5.7 M. This decreases the minimum achieved TPC by \$45 M and \$100 M compared to LR and GA alternatives, respectively.

The normalized values are included in Fig. 10 where the following case studies are depicted: the first case concerns a small power system consisting of 10 generating units supplying a 24-hour power demand, the second case study regards an isolated system with 20x generators and 336 half-hour demand in the presence of RG such as biomass, solar and wind and three further scenarios based on the later. These scenarios consider increased load demand and renewable energy sources with the support of 40, 80 and 100 available generating units, respectively. As can be observed, GA recommends better schedules during the initial function evaluations (below 2000) in both cases. Based on the best population, it generates better function evaluations until the feasible solutions are exhausted (12000). Meanwhile, BO recommends each configuration calculating the maximum expected improvement, considering that there exist feasible solutions which have not been examined yet. This is tuned by the employed acquisition function which leads the optimization also into regions with the higher uncertainty. This justifies the fluctuation before the final, best recommendation of the BO loop. On the contrary, LR starts with a low lambda value which holds the solutions on infeasible and consequently expensive spaces, while in the

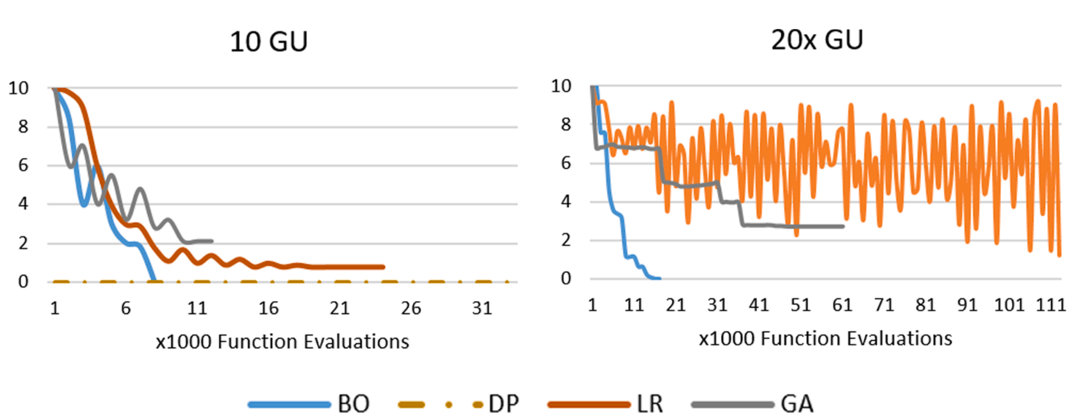


Fig. 10. Potential performance of the considered optimization methods.

case of increased generating units with identical cost-coefficients it falls into unintentional oscillations which compromise the overall optimization performance. We offer the total number of function evaluations required by each method for the test-case studies in Table 6. Their inherent relationship is also demonstrated in Fig. 11.

A weakness of BO may concern the execution time needed to evaluate the objective function. This is due to the computational effort required to inverse the kernel matrix (see Eq. (17)) and the needed time increases by increasing the dimensionality. As a result, when the number of participating generators was greater than 40 and the total intervals accounted for 336 half-hours, the execution time was comparable with that of GA. However, the time horizon for unit commitment is broadly stated up to one week, representing either 168 hourly or 336 half-hourly intervals. Long-term planning (e.g. annual) may involve weekly simulations which can be applied separately or repeated automatically by defining the initial status of all generators and their actual power output as an input to the next week. Consequently, BO may become insufficient in multi-bus formulations which simultaneously consider the optimal placement of storage devices. This would multiply the dimensionality, leading to extreme computational burden and dramatic increase in execution time.

6. Conclusions

In this work, we proposed a novel approach for addressing the unit commitment problem, which is based on the Bayesian optimization paradigm. Two were the main motivational forces behind our work: 1) We sought an effective means of alleviating the need of conventional methods to provide a parametric functional form for the optimized cost function. These are clearly based on rough and simplistic assumptions, which we expected that essentially undermine the effectiveness of the unit commitment solution. 2) We targeted a solution that is intrinsically designed to converge in the least possible number of function evaluations (iterations). Our proposed solution was based on an effective Gaussian process algorithm for enabling data-driven inference of the functional form of the underlying cost function, as well as the utilization of an efficient scheme for selecting the next function evaluation, namely an expected improvement-based acquisition function. We provided an extensive experimental evaluation of our approach under a standard benchmark scenario, and we compared with adopted alternatives. As we observed, our proposed approach completely outperforms the alternatives in terms of the ultimate system costs, as well as the number of required function evaluations. Moreover, the conventional approaches based on complete enumeration, mathematical Lagrange-relaxation frameworks and heuristic Genetic algorithm indicate an exponential increase of iterations in relation to the imposed generating units. Our Bayesian optimization solution presents a linear relationship between the participating generating units and number of iterations required by the optimization procedure which is of utmost importance for large-scale power systems. These findings strongly collaborate our theoretical claims motivating this work. As for future directions for research, we indicate the consolidation of several consecutive hours into one stage. More extended works may also involve multi-bus formulations introducing the real power network losses and transmission constraints into the unit-commitment task. In addition, we urge researchers to consider alternative kernel functions in order to conclude to the best-fit combination in different system networks. We reckon that by increasing the formulation complexity our approach might allow for even higher optimization performance.

CRediT authorship contribution statement

Pavlos Nikolaidis: Conceptualization, Investigation, Methodology, Software, Data curation, Writing - original draft, Visualization. **Sotirios Chatzis:** Conceptualization, Methodology, Supervision.

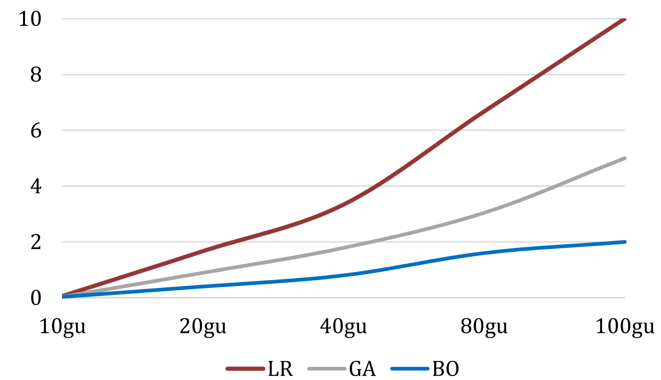


Fig. 11. Total required iterations in relation to the number of the participating generating units.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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